

Lecture 2 - Mathematical Foundations

Contents

- Notation Used in This Course
- Scalars, vectors, and matrices
- Dot products
- Matrix multiplication
- Norms
- Eigenvectors and eigenvalues
- Derivatives
- Partial derivatives
- Gradients
- Practice Problems and Solutions

This lecture reviews the linear algebra and multivariable calculus used throughout the course. An additional introduction to python notebook is available in the Code folder.

Notation Used in This Course

Mathematical symbols do more than name quantities: their capitalization and formatting tell us what kind of object they represent. The following conventions will be used throughout the course.

Object	Notation	Interpretation
Scalar	a, x, θ	A single, fixed number
Vector	$\mathbf{x}$	A fixed column vector with components $x_1, \dots, x_D$
Matrix	A	A fixed rectangular array with entries a_{ij}
Scalar random variable	X	A quantity whose value is not known before it is observed
Realization (or instantiation) of a scalar random variable	x	A particular value taken by X
Random vector	$\mathbf{X}$	A vector whose components $X_1, \dots, X_D$ are random variables
Realization (or instantiation) of a random vector	$\mathbf{x}$	A particular vector value taken by $\mathbf{X}$, with observed components $x_1, \dots, x_D$

Three visual cues are especially important:

- **Bold symbols denote vectors.** Thus x is a scalar, while $\mathbf{x}$ is a vector, and when specified, instantiated vectors for vector RVs.
- **For random quantities, uppercase denotes the unobserved random object.** Thus X is a scalar random variable and $\mathbf{X}$ is a random vector. Uppercase letters such as A and B are also conventionally used for matrices; the surrounding definition and dimensions make their role clear.

- **Lowercase symbols denote realized or fixed values.** Once X has been observed, its realized value is written x ; once $\mathbf{X}$ has been observed, its realized vector is written $\mathbf{x}$.

Random Variables and Their Realizations

A random variable and its realization are related, but they are not the same object. Before a measurement is made, X represents the uncertain quantity. If the observed value is 2.4, we write

$$x = 2.4.$$

The notation

$$X = x$$

describes the event that the random variable X takes the particular value x . For example, $X = 2.4$ connects the random quantity to the value that was observed.

The same distinction applies component by component to a random vector. Suppose temperature and pressure are represented jointly by

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}.$$

Here $\mathbf{X}$ is a random vector, while X_1 and X_2 are scalar random variables. If the observed temperature and pressure are 21.3 and 101.2, respectively, the realization is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 21.3 \\ 101.2 \end{bmatrix}.$$

We may summarize this relationship as

$$\mathbf{X} = \mathbf{x}.$$

When several vector observations are collected, $\mathbf{x}_n$ denotes the n th observed vector, and x_{nd} denotes its d th component. For example,

$$\mathbf{x}_n = \begin{bmatrix} x_{n1} \\ x_{n2} \\ \vdots \\ x_{nD} \end{bmatrix} \in \mathbb{R}^D.$$

The same lowercase bold notation $\mathbf{x}$ is also used for an ordinary deterministic vector that was not generated randomly. In either case, bold lowercase means that the vector is being treated as a concrete, fixed value. Context tells us whether it arose as an observation.

Distribution Notation

When a distribution is written as $p_X(x)$, the subscript X identifies the random variable and the argument x is a possible realization. Similarly,

$$p_{\mathbf{X}}(\mathbf{x})$$

describes a probability mass function or probability density for the random vector $\mathbf{X}$, evaluated at the concrete vector $\mathbf{x}$. When the intended random variable is clear, the subscript is often omitted and these are written simply as $p(x)$ and $p(\mathbf{x})$.

Scalars, vectors, and matrices

A **scalar** is a single number, such as 3, -1.5 , or π . We usually write scalars with lowercase italic letters, such as a or x .

A **vector** is an ordered list of numbers. A vector with n entries belongs to $\mathbb{R}^n$.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

Two vectors can be added when they have the same number of entries, and a vector can be multiplied by a scalar:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

A **matrix** is a rectangular array of numbers. If A has m rows and n columns, then $A \in \mathbb{R}^{m \times n}$:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The entry a_{ij} is in row i and column j . The **transpose** A^T exchanges rows and columns. For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

The $n \times n$ **identity matrix** I_n has ones on its main diagonal and zeros elsewhere. It satisfies $I_n \mathbf{x} = \mathbf{x}$ and $AI_n = A$ whenever the dimensions are compatible.

Dot products

The **dot product** of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

For example,

$$\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = 1(3) + 2(0) + (-1)(4) = -1.$$

The dot product measures alignment. If θ is the angle between two nonzero vectors, then

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta.$$

Vectors are **orthogonal** when their dot product is zero.

Matrix multiplication

The product AB is defined when the number of columns of A equals the number of rows of B . If

$$A \in \mathbb{R}^{m \times n} \quad \text{and} \quad B \in \mathbb{R}^{n \times p},$$

then $AB \in \mathbb{R}^{m \times p}$ and

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Each entry is the dot product of a row of A with a column of B . For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1(5) + 2(6) \\ 3(5) + 4(6) \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Matrix multiplication is associative and distributive, but it is generally **not commutative**: AB does not usually equal BA .

Norms

A **norm** measures the size of a vector. Three common vector norms are

$$\begin{aligned} \|\mathbf{x}\|_1 &= \sum_{i=1}^n |x_i|, \\ \|\mathbf{x}\|_2 &= \sqrt{\sum_{i=1}^n x_i^2}, \\ \|\mathbf{x}\|_\infty &= \max_i |x_i|. \end{aligned}$$

For $\mathbf{x} = [3, -4]^T$,

$$\|\mathbf{x}\|_1 = 7, \quad \|\mathbf{x}\|_2 = 5, \quad \|\mathbf{x}\|_\infty = 4.$$

The distance between vectors $\mathbf{x}$ and $\mathbf{y}$ is often measured by $\|\mathbf{x} - \mathbf{y}\|_2$.

For a matrix $A \in \mathbb{R}^{m \times n}$, the **Frobenius norm** treats all entries as one long vector:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}.$$

Eigenvectors and eigenvalues

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. A nonzero vector $\mathbf{v}$ is an **eigenvector** of A if

$$A\mathbf{v} = \lambda\mathbf{v},$$

where the scalar λ is the corresponding **eigenvalue**. Multiplying by A changes the length or direction of an eigenvector only by the factor λ .

For example,

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The coordinate vectors are eigenvectors because

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus, the eigenvalues are 2 and 3. In general, eigenvalues are found by solving the characteristic equation

$$\det(A - \lambda I) = 0$$

and then solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue. Eigenvectors are not unique: any nonzero scalar multiple of an eigenvector points in the same eigendirection.

Derivatives

For a single-variable function $f(x)$, the **derivative** is the instantaneous rate of change of the output with respect to the input:

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Geometrically, $f'(x)$ is the slope of the tangent line to the graph of f at x . For example, if

$$f(x) = x^3 - 2x,$$

then

$$f'(x) = 3x^2 - 2.$$

Useful rules include

$$\frac{d}{dx}x^n = nx^{n-1}, \quad \frac{d}{dx}(fg) = f'g + fg',$$

and the chain rule

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

Partial derivatives

A function of several variables can change in several independent directions. A **partial derivative** measures the change with respect to one variable while holding the others fixed.

For

$$f(x, y) = x^2y + \sin y,$$

the partial derivatives are

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + \cos y.$$

The symbol ∂ distinguishes a partial derivative from the ordinary derivative d/dx of a single-variable function.

Gradients

The **gradient** collects all first-order partial derivatives of a scalar-valued function into a vector. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

For the previous example,

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + \cos y \end{bmatrix}.$$

At a given point, the gradient points in the direction of steepest increase of f . Its negative, $-\nabla f$, points in the direction of steepest decrease, which is why gradient-based optimization updates often take the form

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k),$$

where $\alpha > 0$ is the step size.

Practice Problems and Solutions

1. Dot Product

Compute the dot product of $[2, -1, 3]^T$ and $[4, 2, 0]^T$.

 Solution

Click to show 

2. Matrix–Vector Multiplication

Multiply $\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$ by $[4, 5]^T$.

 Solution

Click to show 

3. Vector Norms

Find the 1-, 2-, and ∞ -norms of $[-2, 1, 2]^T$.

 Solution

Click to show 

4. Eigenvalues and Eigenvectors

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}.$$

 Solution

Click to show 

5. Partial Derivatives and Gradient

For $f(x, y) = x^2 + 3xy + y^2$, compute both partial derivatives and the gradient.

 Solution

Click to show 